An AI-assisted investigation of a Spin(10) × global $U(1)_{PQ}$ gauge/scalar candidate

Construction, bounded results, and failed tests

Solo research discussion note, prepared with AI assistance. Version 6.

Abstract

This note specifies a nonsupersymmetric Spin(10) gauge/scalar construction with global PQ selection rules. Earlier scalar failures, a separate classical feasibility witness with a large singlet correction, and the failed hierarchy of the older fixed-reference bosonic two-loop calculation remain unchanged. New work constructs further selected one-loop fermion/source normalizations, a bounded local Ward relation and a common family of proper nonminimal-source products. The latest connected-response analysis identifies an unresolved tadpole, nonlinear-mean and joint infrared sewing bridge. Shared-error controls also exposed and corrected a finite-test representation mistake. These are scoped computational and analytic results, not a full connected QAP theorem, massive matter threshold, controlled physical vacuum or unification result. All point, reference and domain restrictions remain; publication makes no new scientific acceptance decision.

1. Assumptions and fields

Work in four spacetime dimensions with gauge group Spin(10) and global PQ selection rules. The simply connected group permits the fermionic spinor 16; local tensor calculations use the $\mathfrak{so}(10)$ action. A possible quotient by common gauge/PQ centers has not been fixed, although the ordinary-field center identification has now been calculated (Section 5). No supersymmetry, explicit soft PQ violation, or additional scalar representation is assumed.

Field	Representation/type	q_{PQ}	Real scalar coordinates
Φ_{ij}	real symmetric traceless 54	0	54
Σ_{ijklm}	complex self-dual 126	+2	252
ϕ_i	complex 10	−2	20
S	complex singlet	−4	2
ψ_p, $p = 1, 2, 3$	left-handed 16_F	+1	—

Indices run from 0 to 9, with Euclidean vector metric δ_{ij} and $\epsilon_{0\cdots9} = +1$. Fix $*\Sigma = +i\Sigma$; its conjugate is the $\overline{126}$. The scalar kinetic convention is

$$\mathcal{L}_{\rm kin} = \frac{1}{2}(D_\mu\Phi_{ij})^*(D^\mu\Phi_{ij}) + \frac{1}{2\,5!}(D_\mu\Sigma_{ijklm})^*(D^\mu\Sigma_{ijklm}) + (D_\mu\phi_i)^*(D^\mu\phi_i) + (\partial_\mu S)^*(\partial^\mu S).$$

The self-dual factor avoids double counting complementary five-index sectors. The physical construction uses the usual gauge-curvature kinetic term and $\mathcal{L} = \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm kin} + \mathcal{L}_Y - V$. The allowed symbolic Yukawa structures are $\psi\psi\phi$ and $\psi\psi\Sigma^*$, with complex symmetric family matrices and isometrically normalized intertwiners. Their numerical matrices and right-handed-neutrino spectrum have not been selected.

The field-content motivation follows Babu and Khan [1]; PQ/axion context is discussed in [2,3]. The potential below is a separately specified source-derived variant. Published scalar benchmarks, fermion fits, and unification conclusions are not imported as results for it. No novelty claim is made.

2. Complete frozen scalar definition

Write $V = V_2 + V_3 + V_4$. Repeated tensor indices are summed over all ten values. Define

$$N = \frac{1}{5!}\Sigma_{abcde}\Sigma^*_{abcde}, \quad u = \phi_i^*\phi_i, \quad t = \phi_i\phi_i, \quad p_n = \mathrm{Tr}(\Phi^n),\ n = 2,3,4,$$
$$K_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma^*_{jabcd}, \qquad T_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma_{jabcd},$$
$$L_{\Phi\Sigma} = \frac{1}{2\,3!}\Phi_{ij}\Phi_{kl}\Sigma_{ikabc}\Sigma^*_{jlabc}, \qquad E = \frac{1}{2!\,2!\,3!}\Sigma_{ijlmn}\Sigma_{ijpqr}\Sigma^*_{lmpqr}\phi_n.$$

T is symmetric traceless. For increasing tuples I, J of length $5-k$, let $P_k[I,J] = \sum_{K\in\binom{\{0,\ldots,9\}}{k}} \Sigma_{KI}\Sigma_{KJ}$ and $Q_k = \sum_{I,J}|P_k[I,J]|^2$, for $k = 0,1,2$. Antisymmetric components with repeated indices vanish. If A, B are disjoint, $\varepsilon(A,B)$ is the parity of their concatenation relative to increasing order; otherwise it is zero. The fourth real five-form quartic is

$$X_{131} = \sum_{I,J\in\binom{\{0,\ldots,9\}}{4}} \sum_{\substack{A\subset I,\ |A|=3\\ B\subset J,\ |B|=1}} \varepsilon(A, I\setminus A)\varepsilon(B, J\setminus B)\varepsilon(A,B)\varepsilon(I\setminus A, J\setminus B)$$
$$\times P_1[I,J]\,\overline{P_1[\mathrm{sort}(A\cup B), \mathrm{sort}((I\setminus A)\cup(J\setminus B))]}.$$

The potential in these normalizations is

$$\begin{aligned}
V_2 &= m_\Phi^2 p_2 + m_\Sigma^2 N + m_\phi^2 u + m_S^2|S|^2,\\
V_3 &= \mu_\Phi p_3 + \mu_{\Phi\phi}\Phi_{ij}\phi_i^*\phi_j + [z_6 t S^* + \text{h.c.}],\\
V_4 &= \lambda_{\Phi1}p_2^2 + \lambda_{\Phi2}p_4 + \lambda_{\Phi\Sigma1}p_2 N + \lambda_{\Phi\Sigma2}L_{\Phi\Sigma}\\
&\quad + \lambda_{\Phi\phi1}p_2 u + \lambda_{\Phi\phi2}\phi_i^*\Phi_{ij}\Phi_{jk}\phi_k + \lambda_{\Phi S}p_2|S|^2\\
&\quad + \lambda_{\Sigma1}Q_0 + \lambda_{\Sigma2}Q_1 + \lambda_{\Sigma3}Q_2 + \lambda_{\Sigma4}X_{131}\\
&\quad + \lambda_{\Sigma\phi1}N u + \lambda_{\Sigma\phi2}\phi_i^*K_{ij}\phi_j + \lambda_{\phi1}u^2 + \lambda_{\phi2}|t|^2\\
&\quad + \lambda_{\Sigma S}N|S|^2 + \lambda_{\phi S}u|S|^2 + \lambda_S|S|^4\\
&\quad + [z_4\Phi_{ij}T_{ij}S + z_K\Phi_{ij}\phi_i\phi_j S^* + z_\eta E + z_D T_{ij}\phi_i\phi_j + \text{h.c.}].
\end{aligned}$$

All m^2, μ, λ coefficients are real; $z_6, z_4, z_K, z_\eta, z_D$ are complex. The masses have dimension two, the cubics dimension one, and the quartics dimension zero. Hermitian conjugation includes the coefficient. The shorthand $\mu_{\Phi\phi}, \lambda_{\phi1,2}, \lambda_{\phi S}, z_\eta$ corresponds to repository names `muPhiPhi`, `lambdaPhiVector1,2`, `lambdaVectorS`, and `zEta`.

There are 29 Hermitian coefficient groups: four quadratic, three cubic, and 22 quartic. Their 24 real and five complex coefficients give 34 real directions before rephasings. Independent D_5 character multiplicities give 34 complex singlet slots before Hermitian pairing; explicit contractions attain that upper bound. The generic scalar phase quotient leaves three relative coupling phases. This basis result fixes the action, not a viable vacuum.

3. Scalar results: a failed tuning and a later positive point

The candidate high-scale background is

$$\langle\Phi\rangle = \frac{\omega}{\sqrt{60}}\,\mathrm{diag}(-2\mathbf{1}_6, 3\mathbf{1}_4), \quad \langle\Sigma\rangle = \frac{\sigma}{4\sqrt{2}}\bigwedge_{k=0}^{4}(e_{2k+1}+ie_{2k}), \quad \langle S\rangle = \frac{v}{\sqrt{2}}, \quad \langle\phi\rangle = 0.$$

The first direction has the intended Pati–Salam/discrete-parity stabilizer; the combined gauge orbit has 33 broken and 12 unbroken directions, consistent with the SM Lie algebra. Global topology and the PQ quotient are separate questions. Tadpoles and Hessians are derived from the same frozen potential.

At the recorded Stage 1 point, $\omega = \sqrt{60}$ in calculation units and $\sigma/\omega = v/\omega = 0.1$. The exact-decimal free coefficients and tadpole-eliminated masses are specified in `STAGE1_POINT.json` and its replay program. All 35 SM-irrep multiplicity blocks were evaluated. The real Hessian has rank 294 and nullity 34: 33 gauge directions plus one PQ direction. The remaining eigenvalues are positive in the canonical kinetic metric; the smallest recorded physical mass squared is approximately 0.08656. This is local positivity modulo symmetry, not global boundedness or a phenomenologically fitted scale. This point has no light Higgs doublet.

The distinct Stage 2 point changes $\mathrm{Re}\,z_6, \mathrm{Re}\,z_K, \mathrm{Re}\,z_\eta$ to 0.3 in the stored dimensionless convention and tunes $m_\phi^2/\omega^2 \simeq -0.1337864654$ to the exact algebraic root recorded in `STAGE2_POINT.json`. It has rank 290 and nullity 38, including four real coordinates of one massless complex SM Higgs doublet. All physical heavy quadratic modes remain positive. This statement alone is insufficient for local stability along the tuned flat direction.

Decisive additional test

For a canonically normalized neutral field h, $H^\dagger H = h^2/2$, expand around the tuned point with physical heavy coordinates X_A:

$$V = V_0 + a_{\mathrm{direct}}h^4 + h^2 j_A X_A + \tfrac{1}{2}X_A\mathcal{H}_{AB}X_B + \cdots .$$

Minimization on the positive heavy subspace gives $X = -\mathcal{H}^{+}jh^2 + \cdots$, and hence

$$\lambda_{\mathrm{eff}} = 4\left(a_{\mathrm{direct}} - \tfrac{1}{2}j^T\mathcal{H}^{+}j\right), \qquad V_{\mathrm{eff}} = \lambda_{\mathrm{eff}}(H^\dagger H)^2 + \cdots .$$

The recorded parent-action evaluation gives

Quantity in $(H^\dagger H)^2$ normalization	Value
Fixed-heavy direct quartic	+0.19910
Heavy relaxation subtraction	56.68893
Relaxed effective quartic	**−56.48984**

The negative sign makes the Stage 2 point a failed local scalar minimum at tree level. It does not invalidate the separate Stage 1 result or prove that every parameter point fails. A direct parent-potential expansion along the heavy-field valley reproduces the quartic; heavy-coordinate shear and complex Higgs-phase controls preserve it. These extraction routes share the frozen tensor implementation. No global-minimum or loop-corrected stabilization claim is made.

A distinct positive-quartic point

The later recorded point in `canonical_so10_positive_higgs/POINT.json` retains the Stage 1 VEVs but uses $z_6/\omega = z_K = 0.03$, $z_\eta = 0.3$, $\lambda_{\Phi\phi 1} = -0.002$ and an exact algebraic tuning $m_\phi^2/\omega^2 \simeq 0.007302047861$. Its real Hessian has rank 290 and nullity 38, with positive heavy modes. The direct quartic is 0.19845235; heavy relaxation subtracts 0.09359411, leaving

$$\boxed{\lambda_{\mathrm{eff}} = +0.10485824.}$$

The point's SHA-256 is `476645438ea8a7ada42fea90e80174b772d45ce3e729d013d134e9848950c816`. Direct-parent and valley-polynomial routes agree with shared tensor inputs. This remains a local tree-level result. Subsequent calculation at this SAME point gives a physical coherent scalar channel with $|a_0| = 0.76809032 > 1/2$ in the fixed-input high-energy tree limit [7]. Its vector-exchange currents vanish in that channel. The point therefore fails this perturbativity screen; positive local masses and quartic did not establish a controlled benchmark. This failure does not exclude the construction. The new exploratory point in Section 6 has different inputs and must be assessed separately.

4. Gauge matching: a calculated mismatch, an incomplete verdict

The recorded direct one-loop calculation matches the full parent to the SM in a common $\overline{\mathrm{MS}}$ background-field Feynman-gauge convention, with GUT-normalized g_1. It includes 290 heavy real scalar directions and 33 massive vector/Goldstone/ghost sets from the later positive-quartic point, not the earlier failed tuning. The threshold ledger binds that point's hash. This point subsequently fails the scalar perturbativity screen. These residuals have not been recomputed for the new exploratory point in Section 6.

Define $\alpha_i = g_i^2/(4\pi)$ and one-loop threshold functions λ_i by

$$\alpha_i^{-1}(\mu) = \alpha_{10}^{-1}(\mu) - \frac{\lambda_i(\mu)}{12\pi}.$$

The complete state accounting gives

$$b_{\mathrm{SM}} = \left(\frac{41}{10}, -\frac{19}{6}, -7\right), \quad b_S = \left(\frac{377}{30}, \frac{77}{6}, \frac{79}{6}\right), \quad T_V = (8, 6, 5),$$
$$b_{\mathrm{SM},i} + b_{S,i} - \frac{7}{2}T_{V,i} = -\frac{34}{3} = b_{10}, \qquad \frac{d\lambda_i}{d\ln\mu} = 6(b_{\mathrm{SM},i} - b_{10}).$$

These are exact beta-jump and one-loop matching-scale checks. At $\mu = \omega$, the frozen inverse-coupling inputs are $(59.0100, 29.5800, 8.47458)$ at M_Z. The one-loop equations are

$$\alpha_i^{-1}(M_Z) = \frac{4\pi}{g_{10}^2} - \frac{\lambda_i^S + \lambda_i^V(g_{10})}{12\pi} + \frac{b_i^{\mathrm{SM}}}{2\pi}\ln\frac{\omega}{M_Z}.$$

They have no common perturbative solution for the frozen inputs. The recorded least-squares residuals, in inverse-coupling units and channel order $(1, 2, 3)$, are:

Completed approximation	r_1	r_2	r_3	$\max\lvert r_i\rvert$
One-loop running and thresholds	0.36615	−1.05980	0.69389	1.05980
Two-loop SM running, one-loop thresholds	0.26091	−0.74548	0.48479	0.74548

The latter uses the SM running framework [5] and frozen reference boundaries from [6], including top, bottom, and tau insertions with their one-loop evolution. A separate inverse-coupling evolution with a different adaptive integrator reproduces the residual to better than 2×10^{-8} under the frozen equations. Matching-scale variation $\mu/\omega \in [0.5, 2]$ leaves maximum residuals from 0.74172 to 0.74931.

What the missing calculation must supply

The complete finite two-loop heavy boundary including matter is still unavailable. All nine retained bosonic primitive families, the prescribed counterterm/light subtraction and the heavy-tadpole bridge have now been assembled in the older point's fixed reference. Their kinetic hierarchy fails at the three tested couplings (Section 8). This is a completed bosonic diagnostic, not a controlled full-matter threshold or a unification refit. The general matching framework [4] supplies methodology and a restricted comparator, not the numerical answer for this construction. Finite matter matching still requires numerical Yukawa matrices and right-handed-neutrino masses.

Small scale variation is not a rigorous bound on unknown finite constants. Setting them to zero or freely tuning them would not compute a prediction. Consequently the formal gauge verdict remains unresolved: neither unification success nor exclusion has been established. The diagnostic fitted $g_{10} \simeq 0.5231$ and $\omega \simeq 1.64 \times 10^{15}\,\mathrm{GeV}$ are not physical benchmark parameters. No replacement spectrum or flavor fit is introduced in this note.

5. Symmetry and quantum evidence: expanded, still scoped

The separately named covariant gauge/scalar candidate satisfies the recorded classical local identities through the 328-real scalar representation and all 45 generators. It is an inactive qualification, not replacement of a promoted component action. Earlier literal-action failures remain preserved; one has

$$\delta\mathcal{L}_{\text{old}} = \frac{ig_{10}}{\sqrt{2}}(\partial_\mu C^4)(Y\partial^\mu X - X\partial^\mu Y) \neq 0.$$

The later candidate repairs that kinetic structure by derivation. A failed encoding does not refute Spin(10). Likewise the preserved auxiliary-cubic normalization counterexample, in common $\hbar g_{10}\xi^2/(32\pi^2\epsilon)$ units, remains

$$G_{\text{actual}} = \frac{48\sqrt{10}}{25}, \quad G_{\text{candidate}} = \frac{8\sqrt{10}}{5}, \quad G_{\text{actual}} - G_{\text{candidate}} = \frac{8\sqrt{10}}{25}.$$

Formal one-loop progress

The earlier scalar-pole closure has rank $4 + 4 + 26 = 34$ within its tested operator space. Subsequent work goes further: the complete formal local one-loop logarithmic residue in the explicitly ordered mixed-reference/heat family obeys $Q\Gamma_{1,\log} = 0$ as an integrated local functional. Its proof retains the source Euler terms, reference subtraction and boundary currents. Cyclicity is used for the integrated homogeneous UV residue, not for an arbitrary finite heat-weighted trace. This is not an identity for the complete finite physical effective action.

The actual parity-even bosonic counterterm has also been reconstructed within 34 real scalar-potential directions, four scalar kinetic metrics and one Yang–Mills metric, plus separate background/vacuum normalizations. These are operator-coordinate counts, not changes to the formal 34/39 authority ledger. Correlated first-order exact-term normalization and physical parameter closure are qualified on the stated coefficient domain. A stricter higher-order extension cannot be imposed retroactively as a missing one-loop coefficient test; the broader original quantum acceptance obligations still remain.

Constructed graded causal products now support a local anomalous quantum action principle at finite order on a compactly switched, off-shell domain [11,12].

The required heat-to-causal finite conversion, simultaneous Slavnov/background normalization and an explicit relative reference member have been retained. This improves on the earlier merely conditional locality discussion. It neither removes the insertion of a classical switching breaking nor proves a global interacting vacuum, physical infrared limit or unique historical prescription. An actual source-quadratic counterexample shows that agreement of relative field products alone cannot determine the full retained-source functional. Full E5/A5 and physical quantum equivalence remain unqualified; the formal state is still version 113, at 34/39.

Gauge anomaly and PQ structure

For the three left-handed 16_F multiplets, the local four-dimensional perturbative gauge-anomaly tensor and linear gauge traces vanish. A weight polynomial route with conjugacy/polarization and a Clifford-matrix route evaluate the established representation criterion [9], including all 16,215 symmetric cubic slots. This does not construct a regulated matter determinant or settle global anomalies, bundles or the full quantum measure.

For the declared PQ charges the convention $N = 2\sum_f q_f T_c(f)$ and $E = 2\sum_f q_f Q_f^2$ gives

$$N = 12, \qquad E = 32, \qquad E/N = 8/3.$$

The common Spin(10) center identifies an ordinary-field PQ quarter-turn; nonzero gauge-singlet S fixes the vacuum-orbit period to $\alpha = \pi/2$. The conventional QCD potential would consequently give $N_{\text{DW}} = 3$, *conditional* on realistic nonzero quark masses/QCD susceptibility, this ordinary orbit and no additional identification or explicit PQ breaking [3]. The electroweak-unbroken reference does not supply that QCD vacuum. This is neither a physical axion mass/abundance prediction nor a full global completion.

6. A new scalar witness and its quantum-control warning

The different rational fixture in `canonical_so10_coherent_joint_witness/POINT.json` uses the same field content, action and VEV shape. Its frozen reduced inputs are $p_1 = 1/100$, $p_2 = 1/50$, $x = 3/25$, $s = 1/1000$, $z = 1/500$, $D = 1/50$, $t = 23/1250$, $\Delta = 0$, $\eta = 1/10000$; the dependent coefficients and larger simple weak root are fixed by the recorded action maps. It is an exploratory fixture, not an adopted physical benchmark. Do not combine its results with the old point's spectrum, thresholds or loops.

Exact specialization of the inherited 12 spectral types (35 blocks, 328 real fields) gives 290 positive directions and 38 symmetry/tuning zeros. The actual weak kernel and all positive neutral relaxation modes give $\lambda_{\rm eff} \simeq 0.1074374161 > 0$ at tree level. This uses a new point-specific light direction, not a transported earlier quartic.

Complete classical coverage at this point

The complete symmetric scalar pair space has 53,956 states. In the contact convention of [7], the recorded structural Gram and block-norm bounds give

$$\|a_0^{\rm contact}\| \le \frac{35439836953}{240000000000\pi} \simeq 0.47003544 < \frac{1}{2}.$$

This is a conservative upper bound, not a largest eigenvalue or complete gauged scattering amplitude. Massless-gauge exchange requires its own infrared-safe treatment; no gauge coupling is silently set to zero.

With $X = \operatorname{Tr}\Phi^2$, $n = N_\Sigma$, $u = \phi^\dagger\phi$, $v = |S|^2$, global invariant inequalities and explicit completion of the complex mixed terms give a rational positive-definite four-norm lower matrix. Its exact minors and a separate rational solver yield, for all 328 real fields,

$$V_4 \ge c(X + n + 2u + 2v)^2, \qquad c = \frac{82440401071117326624733394 9}{17205798939483426705901101207 50} \simeq 4.79143115 \times 10^{-4} > 0.$$

This is global quartic positivity, rather than sampled positive rays. It ensures growth at large fields for this potential. It does not prove that the chosen stationary vacuum is deepest or has a sufficient tunneling lifetime. The old failed points and closed parameter families remain failed in their qualified scopes; these certificates do not transfer to other inputs.

A complete selected one-loop pole coefficient is large

At the same fixture, $Q^2 = \omega^2 = 60$ and the declared $\overline{\rm MS}$ stationary-branch expansion, the isolated singlet has tree squared mass $6/5$. Scalar seagulls and momentum-dependent bubbles, including the physical PQ/Higgs zero modes, give a one-particle-irreducible contribution $-1.36799138 - 0.23557167i$. The required vacuum-source insertion adds $+3.03144395$, so [8]

$$\delta m^2_{S,\rm pole} = 1.66345257 - 0.23557167i, \qquad \frac{|\delta m^2_{S,\rm pole}|}{m^2_{S,\rm tree}} \simeq 1.40004.$$

Field-content and projector checks justify this particular one-loop component. Direct singlet gauge/Yukawa vertices vanish here. The tree singlet plane is isolated and the relevant other tadpole multipliers vanish. This does not make every quantum-vacuum component complete. An exact local RG/log check retains generated Φ–S and Σ–S portals [10] and cancels the scale derivative, but does not reduce the large correction. A small first-order correction is unsupported in this expansion. No reliable physical pole mass/width, resummed solution or universal scheme failure is inferred. Classical feasibility and controlled quantum viability remain distinct.

7. New physical diagnostics and finite-threshold progress

The same singlet has distinct momentum diagnostics

At the exploratory point of Section 6, let $a = m^2_{S,\mathrm{tree}} = 6/5$ and $\Pi(s)$ be its complete retained one-loop self-energy. New combinations give

$$\begin{aligned} E &= \Pi(-a) - 2\Pi(-2a) + \Pi(-3a), & E/a &\simeq -0.01940549, \\ T &= \Pi(a) - 3\Pi(-a) + 2\Pi(-2a), & T/a &\simeq 0.05035452 - 0.19630972i, \\ & & |T|/a &\simeq 0.20266496. \end{aligned}$$

Both cancel every local affine function $c_0 + c_1 s$, including mass and wavefunction subtractions, seagulls and the vacuum-source constant at this order. Their magnitudes are about 1.94% and 20.27% of the tree squared mass. They do not replace or invalidate the earlier 140% local-inclusive pole shift; the absorptive cut remains. Neither a selected small combination nor the large pole shift alone supplies a higher-order error bound.

The absence of a direct S Yukawa vertex at one loop does not close the two-loop singlet problem. An actual scalar/fermion topology has a nonzero unweighted carrier $1.1880059743\,\mathrm{Tr}(Y_{10}^\dagger Y_{10})$ in the recorded units. This is graph support, not an integrated correction. Unfrozen Yukawa matrices and masses remain inputs. Uniform scalar-potential weakening preserves the proven classical properties, but worsens the tested fixed-gauge bosonic radial response; it is not a demonstrated rescue.

Nine bosonic primitive families and their finite assembly

The separate C1 calculation uses the *older* +0.10485824 point, hash `47664543...50c816`. It retains the original $\overline{\mathrm{MS}}$ background-field reference in Feynman gauge, $\xi = 1$. Its subsequent tree perturbativity failure remains. Neither its spectrum nor its two-loop numbers applies to the Section 6 witness.

The retained bosonic families are VVVV, VVV, $VVSS$, VSS, VVS, $SSSS$, SSS, ghost–S and ghost–V. Here S labels any allowed scalar line, rather than only the singlet. All nine have now been assembled with the prescribed parent-minus-family and overall pole subtractions, common light matching and heavy-tadpole bridge. This supersedes the earlier four-family inventory, while retaining the historical partial receipts.

The fixed-reference finite coefficient is an exact-once family sum plus the separately derived stationary heavy-field contribution. The bridge is not inserted twice, and the old incomplete finite pilot is not added again. Formal source-region work also establishes a local bosonic width identity in its declared Euclidean Wilson domain. It does not supply every finite source component, a physical infrared theorem or full E5/A5.

8. The completed bosonic diagnostic fails loop hierarchy

In the original reference set $x = g_{10}^2$ and $L = \log(Q^2/60)$. The assembled two-loop function is $\lambda_{2,i} = F_{0,i} + F_{1,i}L$; its quadratic logarithm cancels by the retained vector-family identity. With the qualified background kinetic normalization,

$$K_i = 1 + k_{1,i} + k_{2,i}, \qquad k_{1,i} = -\frac{x\lambda_{1,i}}{48\pi^2}, \qquad k_{2,i} = -\frac{x\lambda_{2,i}}{768\pi^4}.$$

The three independently evaluated diagnostic couplings give, at $L = 0$:

x	Factor	k_1	k_2
1/2	U(1)	−0.215602	21.210377
	SU(2)	−0.038353	6.138535
	SU(3)	−0.066810	4.909815
3/4	U(1)	−0.269482	34.387749
	SU(2)	−0.017089	12.243740
	SU(3)	−0.066515	9.057263
1	U(1)	−0.308300	49.278335
	SU(2)	0.015471	20.369589
	SU(3)	−0.056806	14.330519

In all nine factor/coupling positions the retained two-loop term exceeds both tree and one loop. This fails the necessary hierarchy screen for these bosonic truncations. Successful arithmetic validation authenticates that negative result; it does not turn the physical screen into a pass. The interior $x = 3/4$ value was calculated from actual family inputs, not interpolated between endpoints.

A narrower scale test keeps all coefficients fixed and changes only their common explicit logarithm. Its minimax results are

$$\min_L \max_i |k_{2,i}(L)| \simeq \begin{cases} 3.20229, & x = 1/2, \\ 4.83315, & x = 3/4, \\ 6.93237, & x = 1. \end{cases}$$

Exact rational interval/pairwise certificates on the frozen decimal tables also exceed one. Thus this particular common-log adjustment cannot make all three retained two-loop corrections smaller than tree. It is not physical RG evolution, which also moves parameters, or a scale-independent exclusion of the theory.

No new scalar point, physical coupling or finite scheme was adopted. Uncomputed matter amplitudes are neither zero nor assigned compensating values. Positive truncated K_i is not convergence; treating it as an exact resummation would insert unsupported higher orders. These results do not exclude every positive coupling, different scalar inputs or every admissible quantum prescription. They specifically prevent presenting the tested fixed-reference bosonic expressions as controlled threshold predictions.

9. Symbolic fermion normalization and a finite comparison

The three 16_F families and the symbolic symmetric 3×3 matrices Y_{10}, Y_{126} are now retained in further one-loop calculations. This is symbolic prerequisite work; no numerical flavor fit or realistic neutrino spectrum has been adopted.

The recorded UV results include fermion/scalar kinetic and mass inputs, proper Yukawa and gauge-vertex families, and correlated matter-source normalizations. A full barred-physical Yukawa parameter-pole assembly gives

$$\delta Y_{R,\mathrm{pole}} = \frac{1}{16\pi^2\epsilon}\left\{\frac{W^T Y_R + Y_R W + k_R Y_R}{4} - \frac{135}{8} g_{10}^2 Y_R\right\}, \quad R = 10, 126,$$
$$W = \frac{5}{8} Y_{10}^\dagger Y_{10} + \frac{63}{8} Y_{126}^\dagger Y_{126}, \qquad k_R = \mathrm{Tr}_{\mathrm{family}}(Y_R^\dagger Y_R).$$

The first leg is transposed, not Hermitian conjugated. The trace already contains the three families. This is a scoped parameter-pole input, not an adopted finite prescription or a complete physical running/matching fit. Independent proper gauge-vertex and source projections agree with their previously fixed kinetic normalization; coefficients were not fitted to a desired cancellation.

What the finite fermion result does and does not say

The closed-loop result assembles all calculated field-dependent, mass-independent, parity-even closed-fermion-loop bosonic finite-comparison families in the declared nonexceptional Euclidean domain. It includes pure-gauge, scalar kinetic, mixed-current/quartic and scalar-potential terms. The full 48-dimensional internal/family trace counts each family once. The relative potential coefficient 5/12 is a calculated comparison term; it has not changed the physical scalar potential.

The local density and its algebraic Ward image are reconstructed with explicit Euler and boundary terms. The actual transpose quotient has 77 density words and 236 Ward words. Agreement of two routes establishes the stated finite-functional identity, *not* a vanishing physical quantum anomaly or a newly integrated breaking insertion. The Ward image itself need not vanish. Three tested invariant directions remain arbitrary; the calculation does not prove that they exhaust the allowed space.

The comparison deliberately excludes the parity-odd chiral trace. Its negative finite functional is a relative counterterm candidate, whose full source, background, external-fermion and massive admissibility remains open. It is not automatically an admissible chiral regulator or an adopted normalization scheme. Subsequent calculations have constructed the physical open-spinor embedding and selected gauge/Yukawa finite kinetic and vertex comparisons, including the required internal hatted-vector contributions. These have their own declared mass-independent Euclidean domain; they do not follow from a closed-trace parity average or prove massive physical admissibility.

Complete finite massive matter thresholds, determinant phase/measure, physical infrared limits and E5/A5 remain unqualified. Earlier failed chirality/interface constructions and corrected normalization/source attempts are preserved. Shared action, inherited loop inputs, software and AI authorship limit independence. These formal results do not repair the old bosonic hierarchy failure or the different witness's singlet warning.

10. Selected local Ward relations and proper source products

The new calculations extend the earlier fermion work to an exact-once selected marginal normalization family, including shared bosonic-source coefficients once. Further vacuum-tangent, probe and intrinsic-source comparisons lead to a complete declared local relative symbol and a bounded first-probe Ward relation. They use the records' unshifted SMA/CSW/EDS action and source convention, rather than replacing the older BFT spectrum or the separate formal E5 construction.

The domain is part of the result

Here u is a constant bookkeeping probe, J_a, J_Λ are nonminimal sources, and Π selects the two recorded local source projections called T and F. The construction is one-loop, Euclidean, at fixed A_0, with Schwartz test fields/sources and positive scalar *reference* masses. Those masses are not a proposed physical spectrum. Source/field variation precedes restriction, UV finite part precedes infrared removal, and only the stated first-u, single-Slavnov response is qualified.

Actual port and invariant-block arguments exclude the additional terms in this particular projection while retaining the original master defect. The complete ordered positive-heat cutoff response is generally nonzero. The qualified local relative relation, including its finite conversion, is

$$\Pi\big(\mathcal{B}C_{\mathrm{rel}} + \mathrm{Cutoff}_{\mathrm{FP}}\big) = 0.$$

It does not assert an unrestricted $\mathcal{B}\Gamma_1 = 0$. Gaussian transpose/relabeling is performed at convergent positive heat before finite-part extraction; arbitrary finite weighted-trace cyclicity is not assumed. The complete earlier loop tensors are authenticated reuse, not new integrations for this relation.

A common proper-product extension

Within the inherited first-u formal domain, a common family of proper nonminimal-source products now covers any finite number of independently smeared source derivatives and their response to one Slavnov variation. The all-order-in-source-count statement follows by induction on ordered inverse derivatives; finite matrix checks corroborate it. It is coefficientwise in a formal Born series, not a convergence theorem.

The direct quadratic-source contact and all inverse-response/cross terms are retained. The allowed local source-pair patterns are J_aJ_a, $J_aJ_\Lambda C$, and $J_\Lambda J_\Lambda CC$. The middle pattern has a loop contact and is not populated in the classical quadratic-source vertex. Power counting of local ambiguities does not set nonlocal products or infrared contacts to zero.

An allowed finite source-quadratic functional A_2 remains symbolic. Changing it changes both the second product and the first insertion away from zero source. First-insertion data at zero source cannot fix it. This is a proper, source-integrable extension, not a complete Ward normalization of every contact, unrestricted mixing classification, multiple independent Slavnov operations, physical K-source completion, global chiral phase or full QAP/E5/A5.

11. The connected-response bridge and shared-error limits

Proper products differentiate the effective action at fixed mean fields. Connected products at fixed ordinary sources also differentiate the mean response. With φ the ordinary field and y_i, y_{ij} its first and second classical source responses, the five-term identity for commuting source slots retains

$$\begin{aligned} W_{1,ij} = & -\Gamma_{1,ij} - \Gamma_{1,i\varphi}[y_j] - \Gamma_{1,j\varphi}[y_i] \\ & - \Gamma_{1,\varphi\varphi}[y_i, y_j] - \Gamma_{1,\varphi}[y_{ij}]. \end{aligned}$$

Odd-source calculations retain the ordered left/right derivative signs specified in the paired proof. The last term contains the tadpole response. It cannot be discarded by setting the ordinary mean or one affine endpoint to zero. Classical source-pair contacts and nonlinear mean responses can survive.

The latest work derives the actual source endpoints before restriction, including their zero complements. Nonlinear bosonic and genuine odd-source finite examples check the five-term identity by direct integration, implicit stationarity/determinant methods and Schur/Berezin routes. They are statistics-sensitive algebra tests, not new Spin(10) loop values. A necessary affine soft bound is established; it is not the complete joint sewn-graph domain.

What is still missing

The actual renormalized auxiliary tadpole has not been shown to obey the required adjoint membership condition. The full nonlinear mean response must be derived using the same fixed-heat stationary map and proper subtractions. The general-Schwartz joint loop/external-momentum sewing domain, including zero-transfer tadpole contractions, is not established. A massless inverse need not preserve Schwartz profiles.

Exact controls give different simultaneous smoothing/IR limits $0, 11/18, 1$ for a zero-momentum distribution. Another control shows that a cutoff-dependent tree response multiplied by a power pole leaves a finite term missed by sewing only after separate subtraction. These examples invalidate shortcuts; they do not prove a nonzero obstruction in the actual model. Neither a tadpole nor a contact baseline has been chosen to force cancellation.

Concrete AI-capability failures are part of the evidence

In a finite source-product fixture, two matrix methods agreed because they shared an incorrect action-to-Hessian index translation. Direct Wick/Berezin integration rejected their common answer. After restoring the ordered left/right derivative convention, complete action reconstruction and the independent test agreed. The actual model used that convention already; no physical input or old loop coefficient was changed. Rejected outputs and failed reviews remain preserved.

One intermediate nullspace was also labelled too strongly as background invariant. It was a restricted BRST coefficient annihilator. A separate actual adjoint calculation establishes a two-dimensional invariant space, but the necessary tadpole membership is still unproved. Correcting that interpretation does not retroactively validate the original label.

These results make the remaining bridge explicit. They do not establish the full connected-product prescription, QAP/measure/global phase, Lorentzian or physical-vacuum limit, finite massive matter C1, quantum control or physical unification. Shared action, analytic premises, data/software and AI authorship continue to limit independence.

12. Evidence and reproduction map

The frozen source checkpoint is

`e08257cd2133031e11b255a1ec17173ddb85628a`.

Source `project_state.json` remains version 113, unchanged by publication. Earlier public notes through Version 5 and all prior source archives/manifests are preserved; the v5 source update extends the chain through handoff V661. The reproduction guide specifies exact assembly and separate historical-policy and external-control namespaces. Original Git history is absent; custody exit 2/seven unresolved references is not PASS.

The following folders are under `calculations/`, except the last row. Read the frozen contract, actual completed receipt, inputs, derivation and successful/failed captures together. Earlier reports retain their chronological limits; later certificates do not rewrite them.

Scope	Folder locator
Action / old points	`canonical_so10_scalar_reconstruction/` `canonical_so10_scalar_benchmark/`
Failed / positive quartics	`canonical_so10_light_higgs_quartic/` `canonical_so10_positive_higgs/`
Old contact failure	`canonical_so10_radial_input_control/`

New fixture / bounds	`canonical_so10_coherent_joint_witness/`
	`canonical_so10_complete_contact_bound/`
	`canonical_so10_finite_full_bfb/`

Singlet pole / RG	`canonical_so10_singlet_pole_control/` `canonical_so10_singlet_pole_rg_check/`

Shape / missing input	`canonical_so10_singlet_momentum_shape/`
	`canonical_so10_singlet_two_loop_input_closure/`
	`canonical_so10_scalar_gauge_homothety/`

Historical gauge fit	`canonical_so10_direct_gauge_next_order/`

Formal UV / closure	`canonical_so10_bv_uv_residue_bridge/`
	`canonical_so10_physical_counterterm_span/`
	`canonical_so10_one_loop_acceptance_reconciliation/`

Local products / QAP	`canonical_so10_graded_causal_qap/`
	`canonical_so10_background_reference_normalization/`
	`canonical_so10_local_cutoff_comparison/`

Primitive amplitudes	`canonical_so10_vector_quartic_ir_matching/`
	`canonical_so10_vector_scalar_quartic/`
	`canonical_so10_scalar_current_vss_assembly/`
	`canonical_so10_ghost_scalar_sunset/`

Gauge / PQ structure	`canonical_so10_local_gauge_anomaly/` `canonical_so10_pq_anomaly_screen/`
Portable complete pilot	`tools/portable_dh_v1/README.md`

The guide's dated update links the completed bosonic assembly, hierarchy and common-scale diagnostics, source-region theorem, symbolic Yukawa/source normalizations, selected finite fermion functional, local Ward relation and proper/connected source-product records. Their actual versioned receipts retain the narrower domains described above.

What readers can rerun

The portable pilot needs Python 3.11+ and the standard library. Its slot and left-derivative methods compute every coefficient of a 99,256-position classical Hessian identity before opening the reference table. It inherits the interpretation of its frozen action projections; it does not reproduce the full quantum model or upstream physical proofs. Both methods were run for the preceding release in an isolated copy; their unchanged code/input bytes and completed captures are reused here.

The package verifier authenticates every released file and archive member; the publication receipt separately lists receipt/input checks, actual replays and omissions. Expensive original Hessian, pole and threshold computations are not repeated for packaging. Some historical harnesses retain absolute paths and read guards; explicit relocation is needed, without editing frozen evidence or bypassing integrity checks.

13. Remaining obligations and AI assistance

No physically viable GUT or experimental agreement is established. The new point has a full classical quartic bound, but the preferred vacuum, tunneling, loop control, hierarchy/naturalness and complete gauged scattering remain open. Full quantum restoration/measure/contact/IR obligations, the complete finite heavy threshold, realistic matter/neutrino inputs, complete axion cosmology and proton-decay phenomenology also remain open. The old gauge residuals are diagnostics at their own frozen inputs and do not predict the new point.

Useful discussion questions are how to define a controlled physical-input expansion retaining the computed singlet cut, and how to finish the remaining massive matter and admissibility layers without double counting. Selected local one-loop relations and fixed-reference bosonic amplitudes must be assessed in their own domains. Missing terms are not assumed to rescue any failed result.

LLMs assisted derivations, code, verification and writing. Scientific direction and acceptance belong to the human author. The records demonstrate bounded symbolic/numerical calculation, structural reductions and cross-checking; they also preserve failed interpolation, assertion, normalization and verification attempts. Corrected agreement cannot validate a shared physical assumption. Shared action/representation inputs, inherited proof caches, some tensor/arithmetic code, Python/SymPy and the same analyst limit independence. Selected work uses NumPy, SciPy, mpmath, CVXPY/Clarabel and Julia. Multiple AI routes are not independent scientists. No external expert verification, whole-model proof-assistant certification or general reliability claim is made.

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